Phone: 050-886-9990 | **Email:** yizharf9@gmail.com | **GitHub:** github.com/yizharf9/coding

Education

B.Sc. Electrical and Computer Engineering | Ben Gurion University *Expected Graduation: February 2027*

- **Specialization:** Computer Engineering (Algorithmics) & Signal Processing and Machine Learning
- **Relevant Coursework:** Digital Signal Processing, Optimization Methods, Control Theory, Systems Architecture.

Technical Skills

- **Programming Languages:** Python (pandas, NumPy), C, Rust, JavaScript, HTML/CSS
- **Machine Learning & Math:** PyTorch, Scikit-learn, SciPy, MATLAB
- **Hardware & Architecture:** ESP32, CISC processor architectures
- **Environments & Tools:** Linux (Ubuntu), Git-Bash, GitHub, VirtualBox, LaTeX
- **Languages:** Hebrew (Native), English (Fluent)

Projects & Research

Quantum Key Distribution via Optical Communication | *Final Project* | Link to Project

- Evaluated the performance of Adaptive Optics and Data Modulations and their direct effects on the Bit Error Rate (BER) across an FSOC channel.
- Conducted detailed research and analysis on the signal fidelity of single-mode versus multi-mode fiber for atmospheric communication links.

Generative Models in 3D Artificial Intelligence | *Academic Project* | [Link to Project] fill in link!!!

- Addressed the 'Janus problem' in 3D generation by implementing a native generative architecture, overcoming the multi-view inconsistency inherent in standard 2D diffusion models.
- Developed a novel alternative to traditional rendering algorithms like Gaussian Splatting to enforce strict 3D awareness and object identity consistency.

Digital Signal Processing (DSP) Applications | *Academic Lab* | link for DSP lab content

- Designed and simulated DSP algorithms utilizing MATLAB to analyze and visualize complex frequency responses.
- Developed and deployed a software bandpass filter onto physical DSP hardware using Code Composer Studio, evaluating real-time system performance.